Now we come to synthetic chemistry, the science of the building of new chemical compounds. Bismuth hydrate will be a good example. It had been concluded by chemical analogies that the production of such a compound was bound to be a distinct possibility. But owing to the extreme difficulty of detecting this gas, all attempts to produce it seem to have failed. However, this project has been crowned by success, thanks to the use of radioactive bismuth as a tracer.

It may be mentioned too that in colloid chemistry room for further applications has been found in connection with the detection of colloidal and crystalloid solutions, as well with the ageing of sols and gels.

VII. ARTIFICIAL RADIOACTIVE SUBSTANCES IN BIOLOGY AND BIOCHEMISTRY

Artificially radioactivated substances have also been used, with great advantage, as tracers in biology. In a living organism, changes are often far slower than in a test-tube. This fact calls primarily for the use of radioactive substances with longer half-lives.

One of the most important applications in this field has been the study of metabolism, carried out by Hevesy and others. Formerly, it was only possible to determine the overall picture of a metabolism by verifying what quantities of a certain substance, introduced into the organism for this specific purpose, were still present in its various individual parts after a certain length of time. However, it was impossible to distinguish this particular substance from the chemically identical one previously present in the organism. This circumstance resulted in considerable uncertainty, particularly as regards the speed of the distribution of the substances introduced among the various organs. But the technique of labelling atoms by their radioactivity quickly eliminated this difficulty. This tracer technique enables us to distinguish the atoms which have been deliberately introduced into the organism from those which were there previously.

For instance, Born, Schramm and Zimmer grew tobacco plants in a nutrient soil containing a substance enriched with radioactive phosphorus, which was absorbed by the plants, since phosphorus is one of substances essential for the maintenance of organic life. The progress and paths of the phosphorus in the plants could be observed, and the points of its strongest concentration could be recognized. The major part of the radioactive phosphorus passed into the uppermost, youngest still growing leaves, while less found its way into the lower leaves, and still less into the fully developed leaves. One leaf, in which the circulation of sap had altogether ceased, absorbed no phosphorus at all. Another study of a similar nature determined the speed of travel of the